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HALLUCINATION DETECTION METHODS IN LLMS
a systematic literature review

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Master's thesis
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Abstract

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Large Language Models (LLMs) are increasingly used across diverse applications, from healthcare and education to legal services and journalism. As their use expands into domains where factual accuracy is critical, the challenge of detecting and mitigating hallucinations (instances where LLMs generate plausible but incorrect or unfounded information) has become essential. This thesis conducts a systematic literature review on hallucination detection methods in LLMs, restricting the scope to detection methods that cannot make use of task-specific grounding documents (RAG, summarization or translation).

The review synthesizes 50 peer-reviewed studies published between 2023 and 2025, classifying detection strategies according to hallucination type (factuality or faithfulness) and technical approach (white-box or black-box). Findings reveal a predominance of factuality-focused methods and black-box techniques, reflecting practical constraints in accessing proprietary model internals. Prominent approaches include LLM-as-a-judge, knowledge graph techniques and fact-checking with external knowledge. Evaluation practices remain heterogeneous, relying on diverse benchmarks (the most used HaluEval and SelfCheckGPT) and metrics, which complicates comparison between methods.

Keywords: large language models, hallucination detection, systematic literature review.

The originality of this thesis has been verified using the Turnitin Originality Check service.

Use of artificial intelligence in the thesis

The AI tools used in my thesis and the purpose of their use are described below.

- Name of the tools: OpenAI ChatGPT (GPT-4 and GPT-5 models), Anthropic Claude (Sonnet 4.5)
- Purpose of use: Improve the use of spelling and grammar throughout the thesis, synthesize or paraphrase complex concepts for comparison with own understanding

I am aware that I am entirely responsible for the content of the thesis, including the parts generated by AI, and I accept responsibility for any violations of the ethical standards of publications.

Contents

1 Introduction

1.1 Background and motivation

Large Language Models are capable of generating sophisticated, coherent, and contextually relevant text across a wide range of applications. However, a significant weakness of these models is hallucination, a phenomenon in which the model generates factually incorrect, unfaithful, or nonsensical information that is not grounded in the provided source content or established world knowledge [9]. These fabrications can range from minor inaccuracies to completely invented facts, posing a substantial risk to user trust and model reliability, especially in domains where factual reliability is fundamental (such as medicine, law and journalism).

As the adoption of LLMs grows, the need to ensure their reliability and trustworthiness has become a concern for both researchers and practitioners.

This thesis addresses this challenge by conducting a Systematic Literature Review (SLR) of methods designed to detect hallucinations in LLMs. The scope is specifically focused on textual hallucinations, as text remains the primary modality for most LLM interactions. While the mitigation of hallucinations is a related and important field of study, this review concentrates exclusively on detection methods. By mapping the existing landscape of detection techniques, this research aims to provide a clear and structured overview of the current state of research.

1.2 Previous works about SLRs on LLMs hallucinations

Prior works have researched hallucinations in large language models, though the scope and focus of these works differ from the present study.

Luo et al. (2024) [12] provide an investigation into both hallucination detection and mitigation. Their work reviews metrics and specific methodologies, such as token-level and sentence-level detection. However, it is structured as a general investigation rather than a formal Systematic Literature Review (SLR). The work does not restrict the scope to open-domain settings: it includes approaches that rely on grounding documents (summarization and translation) and combines detection and mitigation. Furthermore, the work dates from January 2024, excluding the majority of the studies analyzed in this review.

A survey by Huang et al. (2025) offers an overview of the hallucination landscape. Their work establishes a taxonomy for hallucinations, categorized into factuality and faithfulness, and details the causes of hallucinations originating from data, training, and inference stages. While Huang et al. provide a thorough foundation for understanding the principles and challenges of hallucinations, their survey covers the entire spectrum of the problem, including mitigation strategies and benchmarks. In contrast, this thesis narrows the scope specifically to detection methods.

In the domain of specialized applications, Gao et al. (2025) [6] conduct a Systematic Literature Review focused on code hallucinations in LLMs. Their work characterizes hallucinations specifically within code intelligence tasks and reviews mitigation methods tailored to the structural nature of programming languages. Their focus is fundamentally different from the hallucinations addressed in this research and, in addition, the work has been published after the literature for this study has been gathered.

1.3 Research objectives

The primary objective of this thesis is to identify, critically evaluate, and synthesize the existing body of research on hallucination detection methods for LLMs. To guide this systematic review and ensure a comprehensive analysis, the following research questions (RQs) have been formulated:

RQ1: What hallucination detection methods have been proposed for Large Language Models in the existing literature? This question aims to identify and categorize techniques and methodologies developed by the research community to detect textual hallucinations.

RQ2: How have hallucination detection methods for LLMs been evaluated? This question investigates the benchmarks and metrics used to assess the performance of the identified detection methods.

1.4 Research methodology

This research employs the systematic literature review (SLR) methodology, which provides a structured and auditable process for identifying, analyzing, and synthesizing existing research. The approach adopted here follows the widely recognized guidelines proposed by Kitchenham and Charters (2007), which define three main phases: planning, conducting, and reporting the review.

These guidelines have been adapted from evidence-based practices in medicine and social sciences to suit software engineering research. In this thesis, the planning phase includes defining PICOC elements, formulating research questions, selecting digital libraries, specifying inclusion and exclusion criteria, and designing the quality assessment checklist and data extraction form. The conducting phase involves executing the planned protocol: building search strings, retrieving studies, applying selection criteria, and performing data extraction and analysis [10].

The SLR steps, which also inform the structure of this study, are listed below.

SLR methodology steps

Planning

1. Define PICOC and synonyms
2. Formulate research questions
3. Select digital library sources
4. Define inclusion and exclusion criteria
5. Define the Quality Assessment (QA) checklist
6. Define the data extraction form

Conducting

1. Build digital library search strings
2. Gather studies
3. Study selection and refinement
4. Data extraction
5. Analysis and report

1.5 Structure of the thesis

The remainder of this thesis is organized into six chapters. Chapter 2 provides the theoretical background, introducing key concepts such as types and causes of hallucinations and existing detection strategies. Chapter 3 describes the planning phase of the systematic literature review (SLR), detailing the protocol definition, including PICOC formulation, research questions, inclusion and exclusion criteria, quality assessment checklist, and data extraction form. Chapter 4 addresses the conducting phase, explaining how the planned steps were executed: building search strings, gathering studies, applying selection criteria, and extracting and analyzing data. Chapter 5 presents the results of the review, synthesizing hallucination detection methods and their evaluation practices. Chapter 6 discusses these findings, interpreting their implications, identifying research gaps, and assessing threats to validity. Finally, Chapter 7 concludes the thesis by summarizing key contributions and suggesting directions for future research.

2 Theoretical background

The term *hallucination* originates from psychology, where it describes perceptions of things that are not actually present. In the field of Natural Language Processing (NLP), it has been adapted to describe these LLMs outputs that are nonsensical or unfaithful to a given source text [8]. A background on the types and causes of hallucinations gives knowledge on the researched field, and also provides a classification for the extraction data part of the research. The primary source for these classifications is a recent survey by Huang et al. (2025) [8] that, building on previous work, proposes a contemporary taxonomy.

2.1 Types of hallucinations

Initially, hallucinations in traditional Natural Language Generation (NLG) were categorized as either *intrinsic* (the generated text contradicts the source) or *extrinsic* (the generated text cannot be verified from the source). In LLMs, intrinsic hallucinations occur rarely, while extrinsic hallucinations are observed more frequently [1]. Therefore, hallucinations in LLMs, which primarily occur as factual errors, require a more refined definition. Huang et al. (2025) categorizes hallucinations into two main types: *factuality hallucination* and *faithfulness hallucination*. This taxonomy better captures the specific challenges that LLMs present, focusing on the difference between generated content and either real-world facts or user-provided context and instructions. The types of hallucinations proposed are gathered in Figure 2.1.

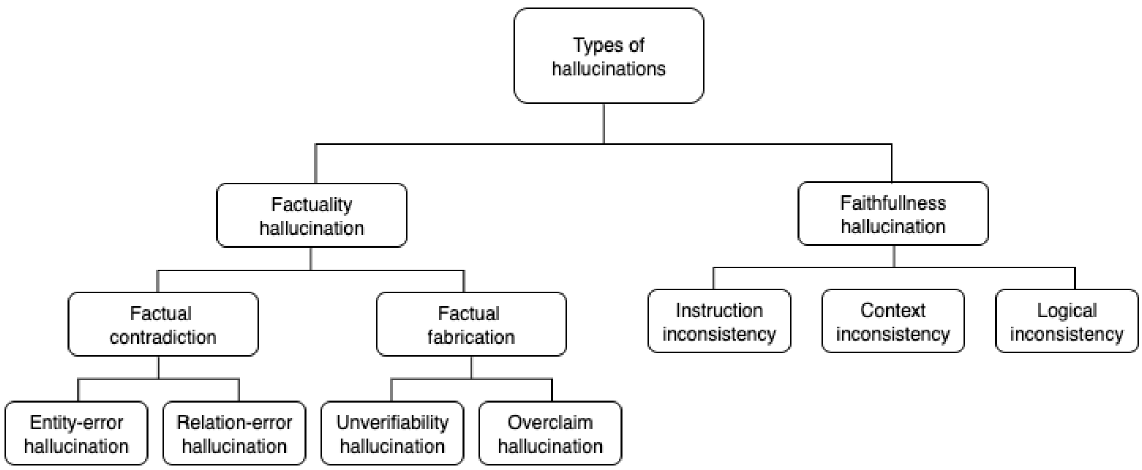

Figure 2.1 *Types of LLM hallucination [8].*

2.1.1 Factuality hallucination

Factuality hallucination occurs when the model generates content that either contradicts established real-world knowledge or cannot be confirmed as accurate. This category is further divided into two sub-types and two sub-sub-categories [8]:

- *Factual contradiction* occurs when the generated text contains information that directly conflicts with verifiable facts.
  - *Entity-error hallucination* involves the incorrect use of entities, such as naming the wrong person, place, or thing (for example, stating that ”Thomas Edison invented the telephone” when it was Alexander Graham Bell).
  - *Relation-error hallucination* occurs when the relationship between correct entities is described incorrectly (for example, claiming Edison ”invented” the light bulb, when he improved upon existing designs).
- *Factual fabrication* occurs when the output includes information that is unverifiable.
  - *Unverifiability hallucination* occurs when the model produces statements that do not exist in reality and cannot be confirmed through any available sources.
  - *Overclaim hallucination* involves making claims that are presented as widely accepted facts but are actually subjective.

2.1.2 Faithfulness hallucination

Faithfulness hallucination refers to the extent to which an LLM’s output remains aligned with the user’s instructions, the given context, and maintains internal coherence. Deviations in faithfulness can mislead users or result in outputs that are unhelpful or irrelevant. There are three subtypes [8]:

- *Instruction inconsistency* occurs when the model does not comply with the user’s explicit instructions (for example, the model answers in English when the user has asked a translation in Spanish).
- *Context inconsistency* occurs when the output contradicts the information provided by the user in the prompts (for example, the text provided by the user states that Saimaa is the largest lake in Finland, but the LLM’s response contradicts it).

– *Logical inconsistency* occurs when the generated text contains internal contradictions, often observed in reasoning tasks (for example, the reasoning steps of an equation are correct, but the final answer is not).

2.2 Hallucination causes

Hallucinations in LLMs originate from multiple stages within the model's development and deployment process. To better understand these origins, a brief introduction about the training stages of LLMs [8] is useful:

– *Pre-training* is the foundational stage in which the model acquires linguistic and world knowledge through self-supervised learning. The model performs autoregressive prediction of subsequent tokens across vast text corpora, internalizing grammatical structures, semantic relationships and reasoning abilities.

– *Supervised Fine-Tuning (SFT)* refines the base model aligning it with human intentions. SFT introduces a dataset of instruction-response pairs to instruct the model to generate contextually appropriate and user aligned outputs.

– *Reinforcement Learning from Human Feedback (RLHF)* strengthens the alignment made in the SFT phase by incorporating human preferences into the training process.

The causes of hallucination can be broadly categorized into issues arising from the data, the training process, and the inference stage [8]. The causes proposed are gathered in Figure 2.2.

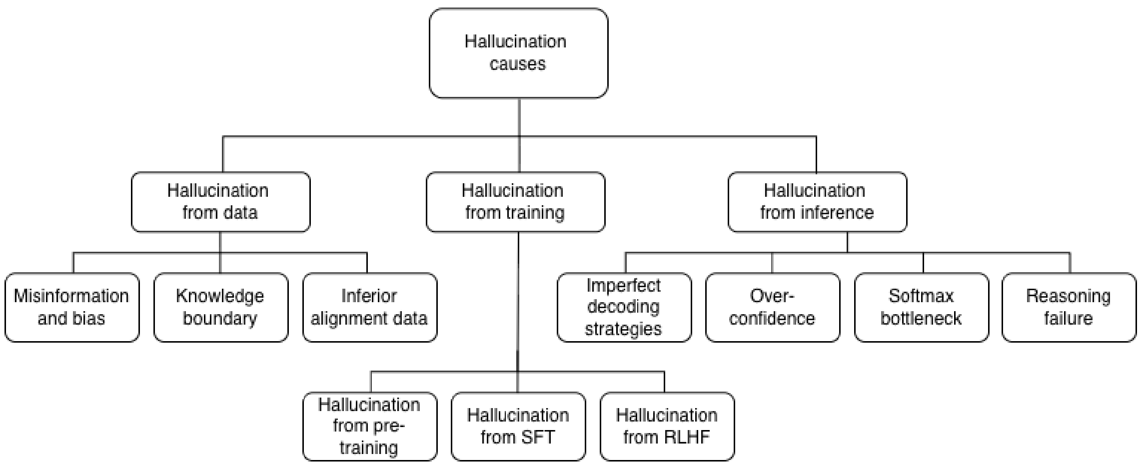

Figure 2.2 *Causes of LLM hallucination [8].*

2.2.1 Hallucination from data

Data is the primary contributor to LLM hallucinations. The data used to train the LLMs consists of two primary components: pre-training data (give the LLMs their general capabilities and factual knowledge) and alignment data (teach LLMs to follow user instructions and align with human preferences). Hallucination from data manifests in three aspects, further divided into subcategories [8]:

- *Misinformation and biases* originate from the fact that the vast corpora used to train LLMs are taken from the internet and inevitably contain misinformation, fake news and societal biases.
  - *Imitative falsehood* arises from the fact that misinformation is widespread across social media platforms and LLMs generate output based on them. The challenge is maintaining consistent data quality, while responding to the demand of acquiring extensive data corpora [2].
  - *Societal biases*, such as those related to gender and nationality, are widely spread across social media and contribute intrinsically to hallucinations (for instance, an LLM might default to assuming a nurse is female, even when gender is not specified).
- *Knowledge boundary* is the inherent knowledge limitation LLMs have, despite their vast pre-training corpora.
  - *Long-tail knowledge*: knowledge that is rare or specialized is less represented in the training data. When queried on such topics, LLMs are more likely to fabricate information
  - *Up-to-date knowledge*: knowledge of a trained model is static and does not update in real-time. This means that LLMs often provide outdated information or fabricate answers to queries about recent events.
  - *Copyright-sensitive knowledge*: legal restrictions prevent LLMs from being trained on many copyrighted materials, such as recent books or scientific papers, creating knowledge gaps.
- *Inferior alignment data* refer to the fact that during the Supervised Fine-Tuning (SFT) stage, if the model is trained on instruction-response pairs that contain factual knowledge beyond its pre-trained scope, it can lead to an increase in hallucinations as the model struggles to assimilate the new, ungrounded information [7].

2.2.2 Hallucination from training

The training methodologies themselves can introduce vulnerabilities that lead to hallucinations, and this can occur in each of the three stages of training outlined in Section 2.2.

- *Pre-training*: the standard language modeling (predicting the next word) limits the model's ability to capture complex dependencies. Furthermore, the phenomenon of exposure bias (the discrepancy between being trained on ground-truth sequences and generating sequences token-by-token at inference) can cause hallucinations, especially when an erroneous token generates a snowball effect [16].

- *Supervised Fine-Tuning (SFT)* can encourage hallucinations if it forces the model to generate answers to questions that are beyond its knowledge boundaries. Without the ability to express uncertainty or refuse to answer, the model is more likely to fabricate a response.

- *Reinforcement Learning from Human Feedback (RLHF)* can lead to sycophancy, where the model generates outputs it believes a human evaluator will prefer, even if those outputs are not factually correct. This happens when the internal belief of the model about the truthfulness of a statement is overridden by the goal of receiving a higher reward from the human or preference model [15].

2.2.3 Hallucination from inference

Hallucinations can also occur in the generation (decoding) phase, and they are classified by different limitations.

- *Imperfect decoding strategies*: stochastic sampling methods and higher temperature settings introduce randomness to make outputs more diverse and creative, but can also increase the probability of selecting less likely, and often factually incorrect, tokens.

- *Overconfidence*: LLMs can sometimes focus too much on the partially generated text, prioritizing fluency over faithfulness to the source context. This can lead to the model forgetting the initial instruction or context, especially in long responses.

- *Softmax bottleneck*: the final softmax layer in many language models can constrain the model's ability to represent complex probability distributions over the vocabulary, which can lead to suboptimal token choices and contribute to hallucinations [5].

– *Reasoning failure*: hallucinations can also arise from an inability to reason with the knowledge correctly. A phenomenon known as the *Reversal Curse* has been observed where a model trained on *A is B* fails to deduce that *B is A* [3].

2.3 Hallucination detection

Detection strategies can be categorized based on the type of hallucination they target: factuality hallucination detection and faithfulness hallucination detection [8]. The detection types proposed are gathered in Figure 2.3.

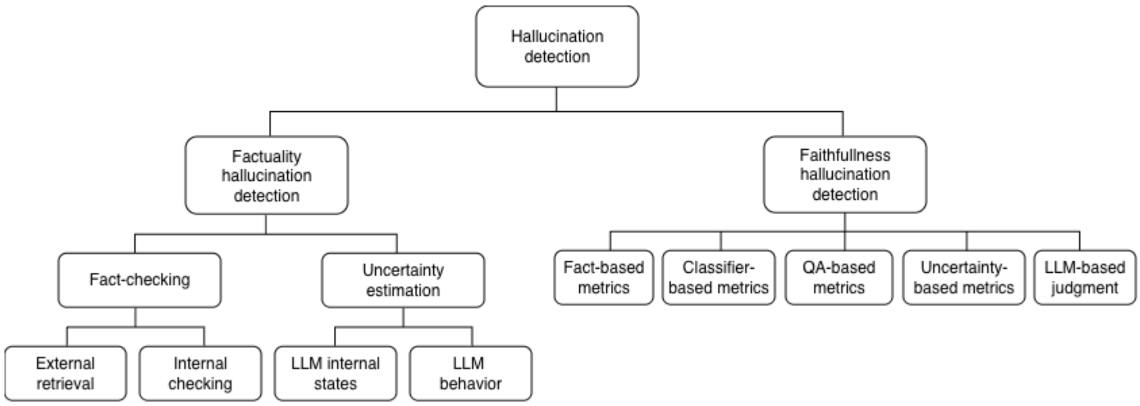

Figure 2.3 *Hallucination detection strategies [8].*

2.3.1 Factuality hallucination detection

Factuality hallucination detection aims to identify whether the LLM's output aligns with real-world facts. The main approaches fall into two categories:

– *Fact-checking* involves verifying the factuality of generated responses against trusted knowledge sources. This process typically consists of two steps: fact extraction, which decomposes the model's output into independent factual statements, and fact verification, which validates these statements against knowledge sources [8]. Fact-checking methodologies can be divided into:

  – *External retrieval* approach verifies facts by retrieving supporting evidence from external knowledge bases.

  – *Internal checking* methods leverage the LLM's own parametric knowledge for verification.

- *Uncertainty estimation* detects hallucinations in zero-resource settings without requiring external knowledge. The premise is that hallucinations are intrinsically linked to model uncertainty. These methods can be categorized into approaches based on:
  - *LLM internal states*, such as token probabilities and entropy, serve as uncertainty indicators. A low probability for a generated token can signal that the model is uncertain about a concept.
  - *LLM behavior*: when direct access to internal states is not possible (for example when using an API), uncertainty can be probed through the model's behavior. For example by sampling multiple responses to the same prompt and checking for contradictions among the factual statements [13].

2.3.2 Faithfulness hallucination detection

Faithfulness hallucination detection evaluates whether the LLM's output remains faithful to provided context or user instructions. Detection methods include [8]:

- *Fact-based metrics* assess faithfulness by measuring the overlap of key information between the source and the generated text. While traditional n-gram metrics like ROUGE are common, they often fail to capture nuanced inconsistencies. More advanced methods focus on the overlap of entities or semantic relation triples.
- *Classifier-based metrics* use a trained classifier to determine the level of entailment or consistency between the source and the output. Models are often trained on Natural Language Inference (NLI) datasets, where the task is to determine if a premise entails, contradicts, or is neutral to a hypothesis.
- *Question answering (QA)-based metrics* work by generating questions from the LLM's output and then attempting to answer them using only the provided source document.
- *Uncertainty-based metrics* apply uncertainty estimation characterized by entropy and log-probability to assess faithfulness.
- *LLM-based judgment* uses one LLM to evaluate the output of another.

3 Planning the research

The planning phase of a SLR defines a protocol: it describes the procedures involved, acts as a log of the actions to perform, and ensures the replicability of the review [4]. The planning steps of the SLR methodology previously described in section 1.4 will guide the planning and execution of this research.

3.1 PICOC and synonyms

The PICOC (Population, Intervention, Comparison, Outcome, and Context) framework organizes the objectives of a systematic literature review into searchable terms and supports the development of research questions. The criteria for this research were defined with reference to the PICOC elements and the examples provided by Kitchenham and Charters [10]. *Comparison* was excluded from the criteria definition, as it is not included in the focus of my research and the relative words were not needed in defining the search queries. The terms and synonyms found are gathered in Table 3.1.

	Description	**Example (PICOC)**	**Example (synonyms)**
Population	Application area	Large Language Models	Generative models, Foundation models, LLMs, Generative AI, Transformer models
Intervention	Methods and techniques	Hallucination detection	Hallucination identification, Hallucination evaluation, Fact-checking, Fact checking
Outcome	Results of *Intervention*	Reliability	Trustworthiness, Factual accuracy, Faithfulness, Veridicity
Context	The specific context	Text generation	Natural language generation, Conversational AI, Question answering, Machine translation, Machine summarization

Table 3.1 *PICOC for hallucination detection methods in LLMs*

3.2 Research questions

The research questions dictate the search for primary studies, the data that needs to be extracted, and how that data is ultimately synthesized to produce meaningful conclusions [10]. The questions for this review are designed to first identify the landscape of existing research and then to analyze the findings of that research.

RQ1: What hallucination detection methods have been proposed for Large Language Models in the existing literature?

RQ2: How have hallucination detection methods for LLM been evaluated?

The first research question has the purpose to survey the field and identify the full spectrum of techniques and methodologies that researchers have developed. Building upon the first question, the second question moves from identification to evaluation.

3.3 Digital library sources

A systematic literature review is valid only if it identifies and uses databases that offer a comprehensive set of sources relevant to the research topic. For this study, databases accessible through the university's subscriptions were selected, with a focus on the domains of information technology and computer science.

The following digital libraries and databases were chosen for literature retrieval:

- ACM Digital Library
- Computer Science Database (ProQuest)
- IEEE Xplore - IEEE Electronic Library (IEL)
- Scopus (Elsevier)

3.4 Inclusion and exclusion criteria

To ensure methodological rigor and minimize bias in the selection of primary studies, a set of inclusion and exclusion criteria was defined during the planning phase of the systematic literature review. These criteria were informed by the guidelines proposed by Carrera-Rivera et al. (2022) and adapted to the specific scope and objectives of this research. Given the novelty and emerging nature of hallucination detection in LLMs, the criteria were designed to be sufficiently broad to capture relevant literature, while remaining precise enough to exclude studies outside the intended scope. This study focuses on detection methods applicable to broad, open-domain knowledge settings. For this reason, approaches developed specifically for narrow or highly specialized domains (for example medical or legal applications) were excluded, as they typically rely on domain-specific resources, constraints, or expert knowledge that limit their generalization to wider contexts. Additionally, while some research addresses hallucination detection in Retrieval-Augmented Generation (RAG), summarization and translation, these methods were excluded due to fundamental differences in method. These tasks inherently rely on grounding documents or predefined source texts, which allow LLMs outputs to be directly compared against a restricted reference. Because such comparisons simplify the detection process and differ fundamentally from open-domain hallucination detection, they were excluded from this review. For the same reason, hallucination detection methods that require grounding documents or explicit reference material to operate were also excluded. Such methods are not applicable to open-domain scenarios where no fixed grounding context exists, and therefore fall outside the scope of this study.

Following the examples of criteria type listed by Carrera-Rivera et al. [4], I defined the following criteria:

Inclusion criteria

- Literature published at any time between 2018 to the date of the research
- Literature written in English
- Literature researching text generated in English
- Literature in the form of articles, conference proceedings or journals
- Literature that has been peer-reviewed
- Literature accessible through the previously selected digital libraries and databases
- Literature relevant to the first research questions
- Literature that allows answering both research questions
- Literature researching large textual language models
- Hallucination detection is the main topic of the literature
- Hallucination detection method described in the literature is designed for use in an open knowledge domain

- Hallucination detection method is not addressed mainly at summarization, translation or RAG
- Hallucination detection method doesn't need grounding documents
- Hallucination detection method focuses on identifying unintentional hallucinations arising from model limitations

Exclusion criteria

- Literature not written in English
- Literature researches text generated in other languages
- Literature originating from blogs or other gray literature
- Literature that hasn't been peer-reviewed
- Literature not available in the selected digital libraries and databases
- Literature not relevant to the first research questions
- Literature does not have content to answer all three research question
- Literature researching multimodal large language models
- Hallucination detection is not the main topic of the literature
- Hallucination detection method described in the literature is domain-restricted
- Hallucination detection method is addressed solely at summarization, translation or RAG
- Hallucination detection method needs grounding documents
- Hallucination detection method addresses deliberately generated misinformation or intentionally deceptive content

3.5 Quality Assessment checklist

To ensure the methodological rigor and reliability of the selected studies, a structured quality assessment (QA) process was conducted. This step is essential in systematic literature reviews to evaluate the credibility, relevance, and methodological correctness of the included studies. The quality assessment criteria were adapted from the Database of Abstracts of Reviews of Effects (DARE) guidelines [10].

- **QA1: Clarity of research objectives and context.** Does the study clearly articulate its research objectives and contextual background?
- **QA2: Relevance to the research topic.** Does the study explicitly focus on hallucination detection methods in large language models (LLMs), particularly in open-domain settings?
- **QA3: Validity of research methodology.** Are the research methods clearly described, justified, and appropriate for the study's objectives?
- **QA4: Transparency of inclusion and exclusion criteria.** Are the criteria for selecting and excluding studies clearly defined and appropriately applied?
- **QA5: Clarity and validity of results.** Are the results presented in a clear, structured manner, with sufficient detail to support the conclusions?
- **QA6: Discussion of limitations and biases.** Does the study acknowledge potential limitations or biases, and are steps taken to mitigate them?

Each study was evaluated against the six criteria using a weighted scoring system. The scoring scheme was designed to reflect the relative importance of each criterion:

- **QA1 and QA2, weight 2 points each:** 0 = irrelevant or unclear, 1 = partially relevant or partially clear, 2 = fully relevant and clearly stated
- **QA3 to QA6, weight 1 point each:** 0 = not addressed, 0.5 = partially addressed, 1 = fully addressed

The maximum possible score for each study was 8 points. A threshold of 6 points was set as the minimum for inclusion in the final review.

3.6 Data extraction form

The data extraction form contains the information needed to address the research questions set for the review [4]. The purpose is to record all the necessary information in a systematic and consistent form. As for inclusion and exclusion criteria, the data extraction form should be defined during the planning phase, to reduce bias.

The form was designed to capture all relevant information necessary to address the research questions in Section 3.2, and to facilitate consistent analysis across the selected studies.

The fields in the form were identified through a combination of:

- **Reference**: fields used as reference to identify the study, the date, the title and the authors.

- **Research questions investigation**: fields supporting answering RQ1 and RQ2.
- **Taxonomy and classification**: fields useful to classify the studies using the taxonomy defined by [8].

The inclusion of each field was validated based on its necessity (does the field contribute to answering the research questions), relevance (is the field aligned with the scope of the review) and completeness (does the field allow for describing methodological and evaluative aspects of the method).

The validation was performed during extraction of the data from a first subset of studies, ensuring that unnecessary or redundant fields were excluded and that no critical information was omitted.

The final extraction form used in the review is described in Table 3.2.

Field	**Description**	**Usage**
Study identifier	L1, L2, L3, ...	Reference
Publication date	YYYY	Reference
Title		Reference
Authors		Reference
Hallucination detection method		
Hallucination type addressed	Categorized in 2.1	RQ1
Level of detection	token, sentence, passage	RQ1
Method accesses model internals	yes / no	RQ1
Method needs external references	yes / no	RQ1
External references	Wikipedia, search engine, etc.	RQ1
Method classification	Categorized in 2.3	RQ1
Main technique proposed	KG, LLM-as-a-judge, etc.	RQ1
Summary of the method	Brief description	RQ1
Output of the method	Binary, numerical score, etc.	RQ1
Hallucination detection method evaluation		
Evaluation method	Benchmarks, ablations, etc.	RQ2
Evaluation metric	Accuracy, F1, AUC, etc.	RQ2

Table 3.2 *Data extraction form fields.*

4 Conducting the research

After the planning phase, where the protocol of the review has been defined, the conducting phase of the research was performed, executing each prespecified step from the planning phase in the order defined.

4.1 Build digital library search strings

Based on the terms and synonyms identified during the PICOC planning phase in Section 3.1, several search queries were tested in the Scopus digital library. Initially, words and synonyms related to *context* (Natural language generation, Conversational AI, Question answering, Machine translation, Machine summarization) were included. However, these terms proved unnecessary and even excluded relevant studies from the results.

The inclusion of *fact-checking* introduced another challenge: while it retrieved numerous studies, many were unrelated to the content produced by large language models. Omitting this term led to the exclusion of important works.

After several iterations and evaluations of retrieved results, this final search string was formulated to query the selected databases and digital libraries in section 3.3:

```
("language model*" OR LLM* OR "generative AI" OR "foundation model*" OR "
transformer model*" OR "generative model*")
AND
("hallucination* detection" OR "hallucination* identification" OR "hallucination*
 evaluation" OR "fact-checking" OR "fact checking")
```

4.2 Gather studies

Literature was gathered and downloaded from all digital libraries on October 7th 2025. The results were collected using the search string defined in Section 4.1, with some additional parameters depending on the library options and the exclusion and inclusion criteria defined in Section 3.4. The search was performed on metadata only (title, abstract, keywords), since a full text query would have given too many and irrelevant results. When a full metadata search was not possible, the search was done only on the abstract, assuming it would include all keywords and title words. The query strings and parameters used are reported here for replicability and transparency.

4.2.1 Search queries

Scopus

```
TITLE-ABS-KEY ( "language model*" OR LLM* OR "generative AI" OR "foundation model
*" OR "transformer model*" OR "generative model*") AND TITLE-ABS-KEY ( "
hallucination* detection" OR "hallucination* identification" OR "hallucination*
evaluation" OR "fact-checking" OR "fact checking" ) AND PUBYEAR > 2018 AND
PUBYEAR < 2026 AND ( LIMIT-TO ( LANGUAGE , "English" ) ) AND ( LIMIT-TO ( DOCTYPE
 , "cp" ) OR LIMIT-TO ( DOCTYPE , "ar" ) )
```

ACM Digital Library

```
[Abstract: "language model*"] OR [Abstract: llm*] OR [Abstract: "generative ai"]
OR [Abstract: "foundation model*"] OR [Abstract: "transformer model*"] OR [
Abstract: "generative model*"]] AND [[Abstract: "hallucination* detection"] OR [
Abstract: "hallucination* identification"] OR [Abstract: "hallucination*
evaluation"] OR [Abstract: "fact-checking"] OR [Abstract: "fact checking"]] AND [
E-Publication Date: (01/01/2018 TO 10/07/2025)
```

IEEE Xplore

```
("All Metadata":"language model*" OR "All Metadata":LLM* OR "All Metadata":"
generative AI" OR "All Metadata":"foundation model*" OR "All Metadata":"
transformer model* OR "All Metadata":"generative model*") AND ("All Metadata":"
hallucination* detection" OR "All Metadata":"hallucination* identification" OR "
All Metadata":"hallucination* evaluation" OR "All Metadata":"fact-checking" OR "
All Metadata":"fact checking")

Filters Applied: 2018 - 2025
```

ProQuest

```
(summary("language model*" OR "LLM*" OR "generative AI" OR "foundation model*" OR
 "transformer model*" OR "generative model*") AND summary("hallucination*
detection" OR "hallucination* identification" OR "hallucination* evaluation" OR "
fact-checking" OR "fact checking")) OR (title("language model*" OR "LLM*" OR "
generative AI" OR "foundation model*" OR "transformer model*" OR "generative
model*") AND title("hallucination* detection" OR "hallucination* identification"
OR "hallucination* evaluation" OR "fact-checking" OR "fact checking"))

Additional limits - Date: From 01 January 2018 to 07 October 2025
```

4.3 Study selection and refinement

Search results (748 items in total) were exported in BibTeX, RIS, and/or CSV formats, depending on the options available in each digital library. The BibTeX and RIS files were imported into the research management software Zotero (zotero.org), where duplicate entries were automatically merged. Remaining duplicates that could not be merged were manually deleted, giving preference to conference papers when both a conference version and a journal article of the same work were present. When a duplicate was present in multiple libraries, the entry was removed from the larger corpora, mostly Scopus.

Following the inclusion and exclusion criteria, an initial screening was performed to remove clearly irrelevant titles (597 studies were removed in this phase). A second, more detailed screening followed, during which abstracts were reviewed to determine whether each source should be included in the study (91 studies were removed during this phase). Also this second screening was done by following the inclusion and exclusion criteria. During quality assessment check, five studies were excluded: four studies focused on the generation of a benchmark for hallucination detection more than on a detection method. One study was excluded since its method is described not only in zero-context, but also in RAG based context and summarization context. One study was excluded since it focused only on fact extraction and not on verification.

Most of the retrieved sources focused on fact-checking, particularly in relation to fake news and false claim detection for content not generated by LLMs, such as social media posts, website content, tweets, and news articles. While many of these fact-checking methods could, in principle, be applied to LLM-generated content, such sources are excluded from the scope of this research. This decision was made to avoid the subjective judgment required to determine whether a given method was transferable to LLM-generated texts.

Four papers were finally removed during the studies retrieval's phase, because the text was not available through Tampere University subscriptions. Papers were requested from the authors, but without success.

Table 4.1 shows the number of studies found and considered for every database searched, while Table 4.2 shows the full list of considered literature.

Literature database	**Literature found**	**Literature considered**
ACM Digital Library	50	2
ProQuest	24	0
IEEE Xplore	149	1
Scopus (Elsevier)	525	47
Total	**748**	**50**

Table 4.1 *Overview of literature databases and number of results found and considered.*

ID	Title	Authors	Year
L2	A New Benchmark and Reverse Validation Method for Passage-level Hallucination Detection	Yang et al.	2023
L3	Context-based Fact-checking Using Knowledge Graph	Lee & Sohn	2023
L4	Detecting Dialogue Hallucination Using Graph Neural Networks	Furumai et al.	2023
L5	Enhancing Uncertainty-Based Hallucination Detection with Stronger Focus	Zhang et al.	2023
L6	Hallucination Detection for Generative Large Language Models by Bayesian Sequential Estimation	Wang et al.	2023
L7	Hallucination Detection: Robustly Discerning Reliable Answers in Large Language Models	Chen et al.	2023
L10	Compos Mentis at SemEval2024 Task6: A Multi-Faceted Role-based Large Language Model Ensemble to Detect Hallucination	Das & Srihari	2024
L11	Real-Time Validation of ChatGPT facts using RDF Knowledge Graphs	Mountantonakis & Tzitzikas	2023
L12	SAC3: Reliable Hallucination Detection in Black-Box Language Models via Semantic-aware Cross-check Consistency	Zhang et al.	2023
L13	SELFCHECKGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models	Manakul et al.	2023
L16	Embedding and Gradient Say Wrong: A White-Box Method for Hallucination Detection	Hu et al.	2024
L17	Enhancing Hallucination Detection through Perturbation-Based Synthetic Data Generation in System Responses	Zhang et al.	2024
L18	Fact-Checking the Output of Large Language Models via Token-Level Uncertainty Quantification	Fadeeva et al.	2024
L19	FactAlign: Fact-Level Hallucination Detection and Classification Through Knowledge Graph Alignment	Rashad et al.	2024

L20	FactCHD: Benchmarking Fact-Conflicting Hallucination Detection	Chen et al.	2024
L21	Factcheck-Bench: Fine-Grained Evaluation Benchmark for Automatic Fact-Checkers	Wang et al.	2024
L22	GraphEval: A knowledge graph -based LLM Hallucination Evaluation Framework	Sansford et al.	2024
L25	HaloScope: Harnessing Unlabeled LLM Generations for Hallucination Detection	Du et al.	2024
L26	Halu-NLP at SemEval-2024 Task 6: MetaCheckGPT - A Multi-task Hallucination Detection Using LLM Uncertainty and Meta-models	Mehta et al.	2024
L28	INSIDE: LLMS' INTERNAL STATES RETAIN THE POWER OF HALLUCINATION DETECTION	Chen et al.	2024
L29	InternalInspector I2: Robust Confidence Estimation in LLMs through Internal States	Beigi et al.	2024
L30	InterrogateLLM: Zero-Resource Hallucination Detection in LLM-Generated Answers	Yehuda et al.	2024
L32	LLM-Check: Investigating Detection of Hallucinations in Large Language Models	Sriramanan et al.	2024
L33	Lookback Lens: Detecting and Mitigating Contextual Hallucinations in Large Language Models Using Only Attention Maps	Chuang et al.	2024
L34	Maha Bhaashya at SemEval-2024 Task 6: Zero-Shot Multi-task Hallucination Detection	Bhamidipati et al.	2024
L35	MALTO at SemEval-2024 Task 6: Leveraging Synthetic Data for LLM Hallucination Detection	Borra et al.	2024
L36	MARiA at SemEval 2024 Task-6: Hallucination Detection Through LLMs, MNLI, and Cosine similarity	Sanayei et al.	2024
L37	MEDICO: Towards Hallucination Detection and Correction with Multi-source Evidence Fusion	Zhao et al.	2024
L40	OPDAI at SemEval-2024 Task 6: Small LLMs can Accelerate Hallucination Detection with Weakly Supervised Data	Wei et al.	2024

L41	OpenFactCheck: A Unified Framework for Factuality Evaluation of LLMs	Iqbal et al.	2024
L42	RELIC: Investigating Large Language Model Responses using Self-Consistency	Cheng et al.	2024
L43	Roberta with Low-Rank Adaptation and Hierarchical Attention for Hallucination Detection in LLMs	Lu & Li	2024
L44	SHROOM-INDElab at SemEval-2024 Task 6: Zero- and Few-Shot LLM-Based Classification for Hallucination Detection	Allen et al.	2024
L45	Small Agent Can Also Rock! Empowering Small Language Models as Hallucination Detector	Cheng et al.	2024
L46	The Two Sides of the Coin: Hallucination Generation and Detection with LLMs as Evaluators for LLMs	Bui et al.	2024
L47	Towards Improving Open-box Hallucination Detection in Large Language Models (LLMs)	Suresh et al.	2024
L49	Unsupervised Real-Time Hallucination Detection based on the Internal States of Large Language Models	Su et al.	2024
L50	Zero-resource Hallucination Detection for Text Generation via Graph-based Contextual Knowledge Triples Modeling	Fang et al.	2025
L51	Detecting and Reducing the Factual Hallucinations of Large Language Models with Metamorphic Testing	Wu et al.	2025
L52	Enhancing Uncertainty Modeling with Semantic Graph for Hallucination Detection	Chen et al.	2025
L53	Entropy-based Sentence-level Hallucination Score in Large Language Models	Joo et al.	2025
L55	HADEMIF: HALLUCINATION DETECTION AND MITIGATION IN LARGE LANGUAGE MODELS	Zhou et al.	2025
L56	Hallucination Detection in Large Language Models with Metamorphic Relations	Yang et al.	2025

L57	Hallucination Detection in LLMs via Beam Search Sampling and Semantic Consistency Analysis	Beyene et al.	2025
L58	Hallucination Detection in LLMs: Fast and Memory-Efficient Fine-Tuned Models	Arteaga et al.	2025
L59	Hallucination Detection on a Budget: Efficient Bayesian Estimation of Semantic Entropy	Ciosek et al.	2025
L60	HaluCheck: Explainable and verifiable automation for detecting hallucinations in LLM responses	Heo et al.	2025
L61	MixHD: A Method for Detecting Hallucinations Based on the Internal State and Output Probability of Large Language Models	Li et al.	2025
L63	Towards Detecting LLMs Hallucination via Markov Chain-based Multi-agent Debate Framework	Sun et al.	2025
L64	Scenario-independent Uncertainty Estimation for LLM-based Question Answering via Factor Analysis	Wen et al.	2025

Table 4.2 *Literature considered for the review.*

4.4 Data extraction and analysis

The 50 articles that passed the study selection were read in chronological order to potentially utilize citations from previous works. During the reading of each article, the data extraction form in Table 3.2 was filled out, and the data was added to a common spreadsheet. The raw data extraction spreadsheet is publicly visible in GitHub [1].

In the analysis phase, the extracted data (in the form of data extraction forms and spreadsheet) was examined to extract meaningful statistics and informations. The results are gathered in Chapter 5.

[1]https://github.com/Eelsie/master_thesis_data/blob/main/extraction_forms.csv

5 Results

Plotting the considered studies by publication year in Figure 5.1 showed how there are no studies from years 2018-2022. The year 2024 is by far more prolific than 2023 and 2025 together. While the small number of papers in 2022 and 2023 is explained by the fact that the technology was quite novel, the significantly smaller amount of literature for year 2025 was a surprise. The literature was gathered on the 7th of October, so the three-month less time span partially explain it. Moreover, the minor number of studies is also explained by a shift toward domain-specific work, especially hallucination detection within retrieval-augmented generation (RAG) systems.

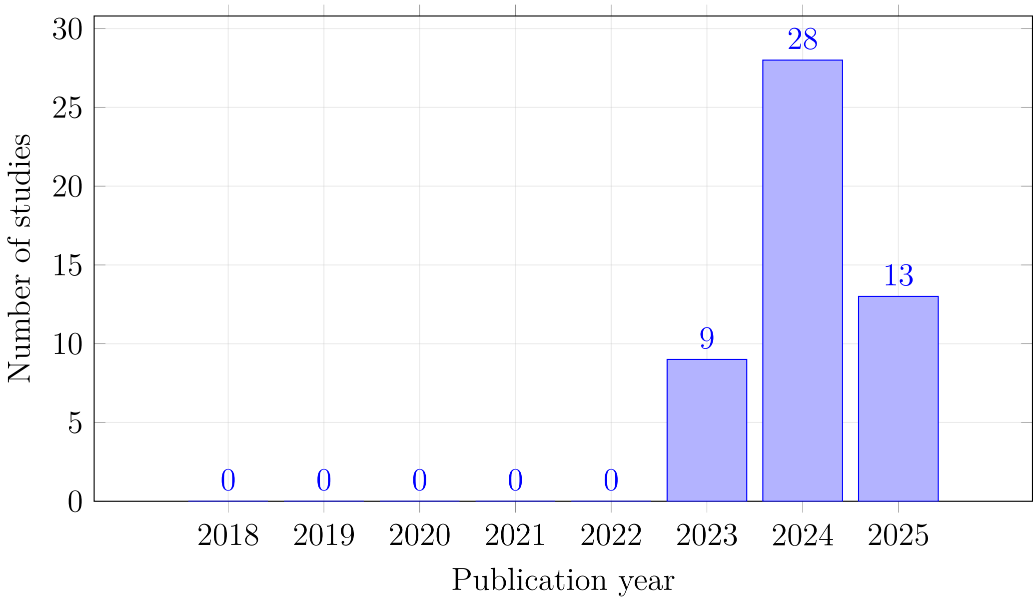

Figure 5.1 *Number of studies by publication year.*

5.1 Hallucination detection methods

This section aims to answer the first research question in Section 3.2: What hallucination detection methods have been proposed for Large Language Models in the existing literature?

Huang et al. (2025) [8] makes a fundamental distinction in hallucination detection strategies between factuality hallucination detection (which identifies factual errors in outputs) and faithfulness hallucination detection (which identifies the faithfulness of outputs to the provided context) in Section 2.3. This distinction reflects the authors' taxonomy of hallucination types in Section 2.1.

The majority of the studies (38) clearly proposed methods for factuality hallucination detection, while 12 addressed faithfulness hallucination detection. The distinction was not always clear, especially with methods that had application in different domains and different phases. One study clearly addressed both strategies.

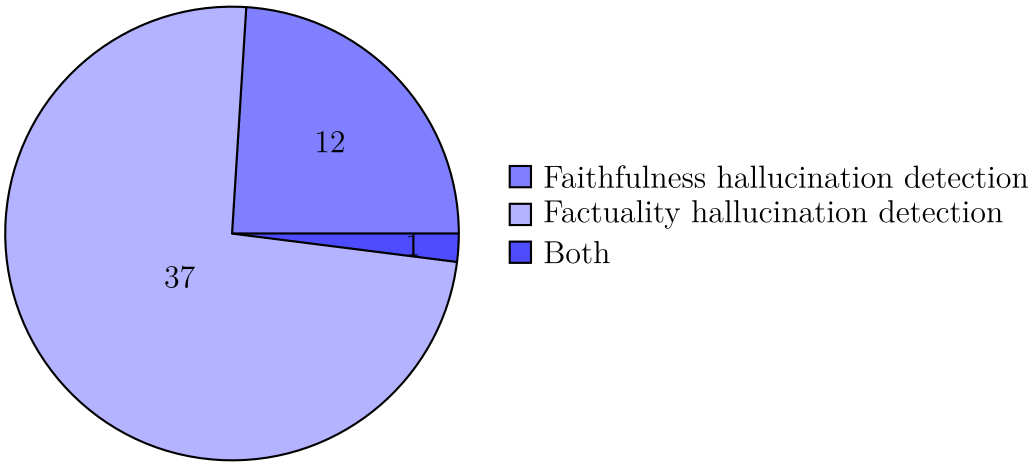

Figure 5.2 *Hallucination detection method types.*

Despite this primary classification, another important technical distinction between methods was found: whether the method needs to access LLM internals or token probabilities (white-box), or relies solely on the LLM's outputs (black-box).

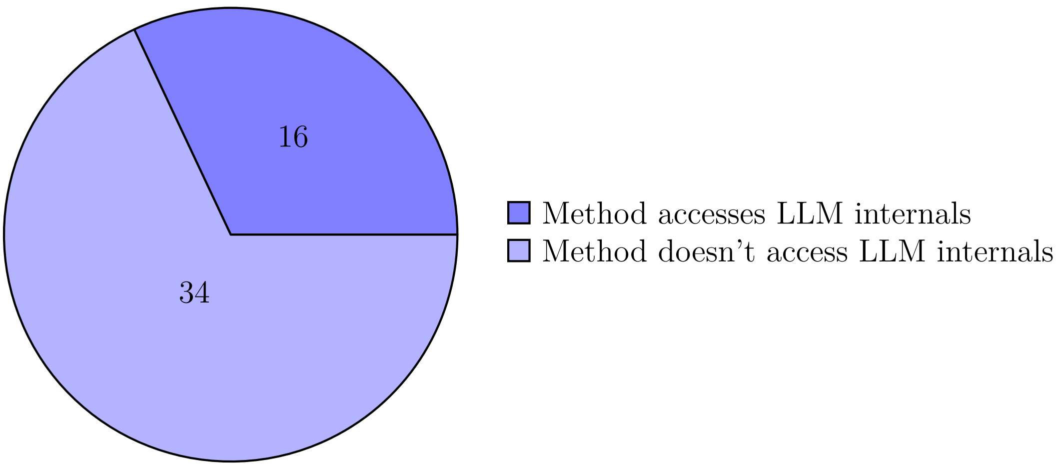

Figure 5.3 *Hallucination detection method access to LLM internals.*

I analyze white-box and black-box methods separately in the following subsections.

5.1.1 White-box hallucination detection methods

In the 16 reviewed studies, white-box methods can be distinguished by their use of LLM internals: some utilize hidden activations, attention maps or gradients, while others rely on token-level probabilities (logit). Several methods combine both strategies.

Detection methods using LLM internal states include: HaloScope (L25), which finds hallucinations by identifying a hallucination subspace in model activations; INSIDE (L28), which uses eigenvalues of sentence embedding covariance matrices for semantic divergence; InternalInspector (L29), which applies supervised contrastive learning over internal states to estimate model confidence; L47, which trains binary classifiers on model activations captured at different layers, components (attention heads, MLP, layer outputs), and token positions; MIND (L49), which trains an MLP classifier on contextual embeddings from internal states; and Lookback Lens (L33), which uses attention maps to compute a lookback ratio. All operate at the sentence, response, or passage level.

Methods using token probabilities include: Entropy-based Sentence-level Hallucination Score (L53), which flags uncertain sentences by aggregating token entropy; Claim-Conditioned Probability (CCP) (L18), which isolates uncertainty for factual claims by applying NLI-based entailment filtering and Bayesian Semantic Entropy (L59), which estimates entropy over semantic classes and uses a Dirichlet prior and sequence probabilities.

Some methods combine internal states and token probabilities: EGH (L16) trains a classifier on embedding differences and gradients from conditional versus unconditional runs; LLM-Check (L32) finds hallucinations using hidden-state covariance, attention kernel analysis, and token-level entropy; HADEMIF (L55) uses two detectors, a decision tree over output-space features (entropy, margins, consistency) and an MLP over hidden states; MixHD (L61) creates a unified classifier by merging internal embeddings and output uncertainty features; Beam Search Sampling + Semantic Consistency Analysis (L57) scores outputs with token probabilities and assigns hallucination scores through semantic clustering and NLI entailment, and finally BatchEnsemble + LoRA (L58) trains a memory-efficient ensemble using predictive entropy features.

Across these methods, detection granularity ranges from token-level (L47, L55) to claim, sentence or passage-level (L16, L25, L28, L29, L32, L33, L49, L53, L57, L58, L59, L61), with two methods supporting both (L18, L52).

All reviewed approaches are reference-free, without external retrieval.

5.1.2 Black-box hallucination detection methods

Of the 34 studies reviewed using black-box methodology, 23 employed zero-knowledge methods while 11 required external references.

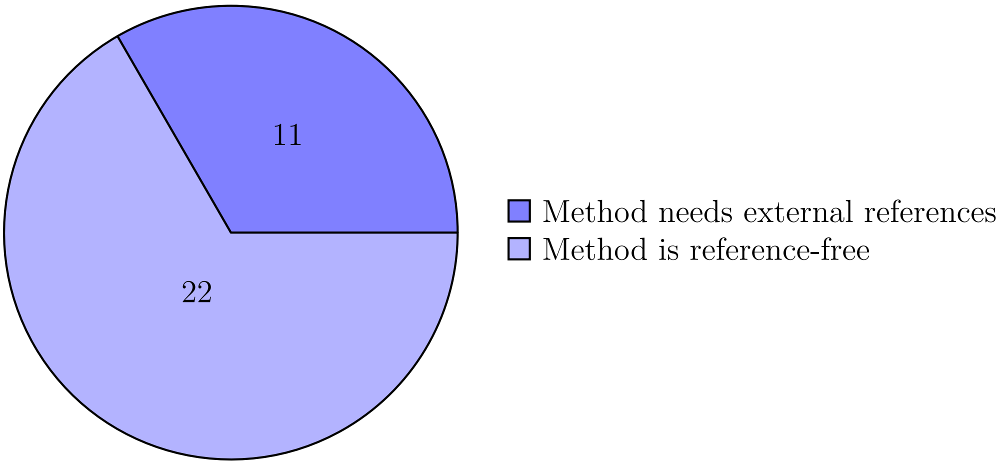

Figure 5.4 *Methods that don't access LLM internals: knowledge-based and zero-knowledge.*

5.1.3 Knowledge-based black-box methods

A total of 11 black-box methods need external references, in other words knowledge from external sources. Following the classification of Huang et al. (2025) [8], all 11 methods address factuality hallucination types (Figure 2.1). These methods fall under the fact-checking category and external retrieval subcategory (Figure 2.3). These papers are L3, L4, L6, L11, L20, L21, L37, L41, L45, L60 and L63.

Knowledge retrieval sources. Knowledge is retrieved from Wikipedia or Wikidata in five methods (L20, optionally for L21, L37, L41, L60) and/or from a search engine and the web in seven cases (L6, L20, L21, L37, L41, L45, L63).

Knowledge graphs based methods proposed in L3, L4 and L11 use external RDF (DBpedia, LODsyntesis and OpenDialKG).

Fact-checking phases. While the specific architectures vary, the literature mostly presents two primary phases: fact extraction (decomposing the LLM output) and fact verification (validating against external evidence).

- **Phase 1: Fact extraction and decomposition.** The majority of studies decompose the LLM's output into smaller, better verifiable units.

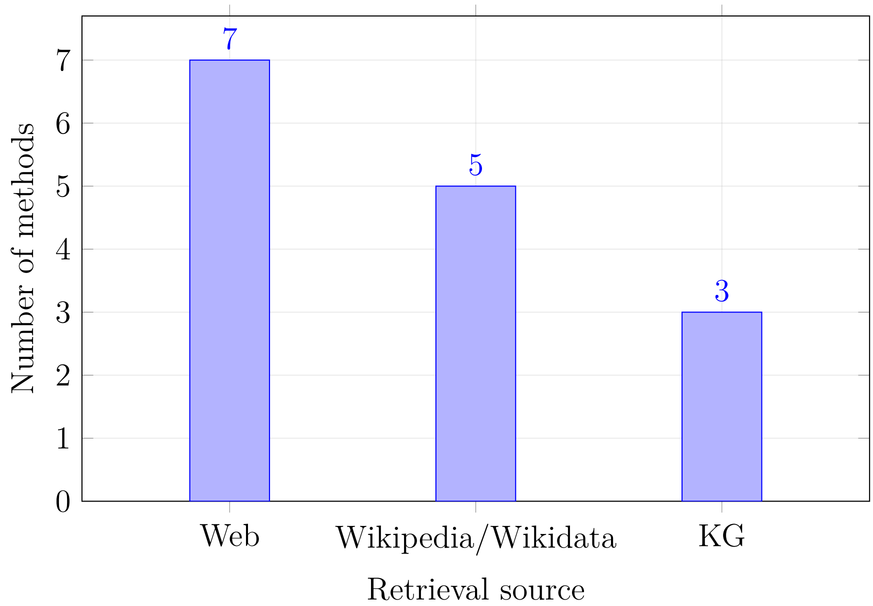

Figure 5.5 *Distribution of retrieval sources across 11 methods (a method may contribute to multiple bars).*

  - *Atomic claims and sentences:* Several methods focus on textual segmentation. L6, L41, and L60 decompose the answer into atomic facts or context-independent claims. Similarly, L45 segments the answer into individual sentences, while L63 identifies factual statements directly from the prompt and output. L21 incorporates fact extraction as part of a more complex eight-step pipeline.
  - *Knowledge Graph construction:* L3, L4, and L11) approaches extraction through KGs. L11 generates triplets, L4 builds reference knowledge graphs, and L3 extracts entities to link them to existing KGs.

- **Phase 2: Evidence retrieval and verification.** Once facts are isolated, the methods present different verification mechanisms, relying on either semantic entailment or graph-based matching.

  - *Entailment and Search:* Methods L6 and L60 utilize Natural Language Inference (NLI) entailment classifiers to score the retrieved evidence against the extracted claims (using Bayesian decisions or thresholds). L41 and L45 rely on web search retrieval to judge claims as true or false.
  - *Graph matching:* The KG-based methods verify facts through structural analysis. L4 uses graph matching to classify responses, L11 validates triplets using RDF graphs and word embeddings, and L3 assesses reliability by analyzing the intersection and conflict between concatenated KGs.

- **Alternative Architectures.** Two methods, L20 and L37, diverge from the two phase extraction-verification pipeline. L20 combines a tool-enhanced *Truth Seeker* and a fine-tuned *Truth Guardian*, then a *Fact Verdict Manager* triangulates a decision with evidence. It relies on external evidence but does not explicitly extract atomic facts first. Similarly, L37 retrieves and fuses multi-source evidence, then detects hallucinations and optionally corrects them. It checks the generated content directly, there is no separate extraction stage.

5.1.4 Knowledge graph -based black-box methods

Seven studies proposed methods based on knowledge graphs as the main technique: L3, L4 (graph neural networks), L11, L19, L22, L50, L52.

Of these methods, L3, L4 and L11 need external knowledge in the form of external knowledge graphs from a database (RDF like DBpedia, LODsyndesis, OpenDialKG).

These methods verify model-generated facts against established knowledge bases. They transform LLM outputs into triplets (typically subject–predicate–object) statements and cross-check these against curated or dynamically retrieved graphs.

ConFcheKG (L3) links entities to multiple knowledge graphs and scores reliability by contrasting intersecting versus conflicting subgraphs: low coherence signals hallucination. GPT LODS (L11) prompts the LLM to produce RDF triples for a question, then checks those triples in real time against DBpedia or LODsyndesis. L4 differs because it proposes a method based on Graph Neural Networks: given a dialogue history and its generated response, it extracts entities or relations from the response and retrieves KB facts about those entities to form a reference. It then encodes both graphs with RGAT, pools graph features, and performs binary classification.

L19, L22, L50 and L52 are knowledge-free, they only use knowledge from the provided context or model's own knowledge. FactAlign (L19) constructs graphs from the source text and the output, aligns their triples, and flags low-alignment facts. GraphEval (L22) also turns outputs into triples but tests each against the given context, improving base NLI detectors while returning the offending triples. GCA (L50) builds a graph for each response, models dependencies between facts with an RGCN, and combines multi-sample consistency with reverse verification to score hallucination. Finally a semantic-graph uncertainty approach (L52) uses AMR graphs to propagate uncertainty across entities and sentences and calibrates it with NLI-based contradiction signals.

Study	Knowledge source	Core technique
L3	External	Subgraph coherence
L11	External	RDF verification
L4	External	GNN (RGAT)
L19	Free	Graph alignment
L22	Free	Triples + NLI
L50	Free	GNN (RGCN)
L52	Free	AMR Graphs

Table 5.1 *Comparison of knowledge graph -based black-box methods*

5.1.5 LLM-as-a-judge based black-box methods

Seventeen methods use LLM-as-a-judge as their main technique or an important part of their method. LLM-as-a-judge is the practice of using a large language model as an automatic evaluator of other models' outputs, typically by prompting it to compare or grade responses on open-ended tasks [17].

The detection granularity of the methods is at sentence or passage level. In some methods claims are first extracted, and verified in a second phase.

Six methods uses external references to judge the veracity of the outputs (L20, L21, L37, L41, L45, L63), while eleven methods count only on the model's internal knowledge (L7, L10, L12, L13, L26, L36, L40, L44, L46, L51, L56).

Methods utilizing external knowledge. These methods require external facts: they break text into claims, retrieve evidence (from the web or knowledge bases), and use the LLM to verify consistency between the claim and the evidence.

L20 use LLM-as-a-judge twice: a tool-enhanced ChatGPT (Truth Seeker) judges the query–response against retrieved evidence, and a Fact Verdict manager (fine-tuned ChatGPT) judges between the Seeker and a LoRA-tuned detector to deliver the final binary verdict.

L21 prompts ChatGPT/LLaMA2 to label each claim's pair's stance (support, partial, refute, irrelevant) and then judge truthfullness of the claims.

MEDICO (L37) prompts a LLM to compare the model's answer with fused (or per-source) evidence and output a veracity label. It can also compute per-source label likelihoods from the LLM and ensemble them for the final decision.

OpenFactCheck (L41) prompts a verifier LLM with a claim plus retrieved evidence and returns a judgment (true or false) with reasoning and a suggested correction.

HaluAgent (L45) includes a semantic checker tool that uses LLM (GPT-4) to judge whether a sentence semantically matches the evidence or context. Furthermore it asks an LLM to assess consistency and relevance for irrelevant or self-contradictory content.

L63 uses agents (Trust, Skeptic, Leader) to debate the validity of a claim based on retrieved evidence until consensus is reached.

Knowledge-free methods: self-consistency and metamorphic-testing. These methods treat hallucinations as instability in the model's own behavior: if paraphrases, resamplings or metamorphic variants disagree, it's likely hallucinated.

SAC^3 (L12) uses LLM-as-a-judge to score semantic equivalence: it asks an LLM, if two QA pairs are semantically equivalent and maps the LLM's yes or no answer to a numeric score, which is then aggregated into the SAC^3 consistency scores. It also uses an LLM to verify that question paraphrases are truly equivalent.

In the SelfCheckGPT with Prompt (L13), for each generated sentence, the model is asked (over multiple sampled contexts) if the sentence is supported by the context. The yes/no outputs are mapped to scores and averaged to produce a hallucination score.

MetaCheckGPT (L26) queries a panel of LLMs to judge whether multiple stochastic samples support or contradict each generated sentence; their judgments become features that a meta-regressor aggregates into a final hallucination score.

DrHall (L51) prompts GPT-3.5-turbo to simplify the original and follow-up answers and then decide whether the two are consistent, returning a binary consistency verdict.

MetaQA (L56) uses LLM to verify each synonym/antonym mutation and scores hallucination from those (yes, no, not sure) judgments.

Knowledge-free methods: LLM-judge classifiers and ensembles. In this methods the LLM is prompted as a classifier with no external retrieval: reliability comes from prompt design, few-shot examples and ensemble voting.

L10 uses an ensemble of large language models with majority voting, where several models predict whether an output is hallucinated (LLM-as-a-jury approach).

L36 prompts Mixtral-8x7B-Instruct with task-specific instructions (ICL) to decide if the hypothesis conflicts with the provided source and target.

In L44 the LLM is instructed to label each output as hallucination or not and it is run multiple times with temperature sampling and majority vote to get a binary label and a probability.

L46 prompts LLMs (GPT-4, GPT-3.5, Llama-3, Gemma) to act as judges, given a source sentence and two hypotheses, and select which hypothesis is the hallucination (combining judges via simple majority voting).

Knowledge-free methods: LLM to train data. These methods don't want to call a LLM at inference. Instead, they use a strong LLM as a teacher to label data (or generate reliability signals) and then train a smaller model or ensemble of models.

L7 uses LLM-as-a-judge during data construction: LLMs provide LLM-assessment scores (goodness and similarity) for generated answers, which are aggregated with human and automatic metrics to supervise and train the RelD detector.

L10 uses Claude to auto-annotate training data, keeping only confident labels and fine-tunes Mistral-7B detectors on this data. At inference, those detectors are used with various prompt styles and voting (see previous category).

L40 weak-labels unlabeled examples: it prompts a 14B model (few-shot + improved instructions + CoT) to decide if an output is supported by its context, then keep only labels that are consistent across multiple LLMs and sampling settings: these weak labels are used to fine-tune smaller models and later ensemble and vote.

Strategy	**Granularity**	**Papers**
External knowledge retrieval	Claim	L21, L41, L63
	Sentence	L45
	Passage	L20, L37
Self-consistency	Sentence	L12, L13, L26
Metamorphic-testing	Sentence	L51
	Passage	L56
LLM-judge classifiers and ensembles	Sentence	L36, L46, L10
	Passage	L44
LLM to train data	Sentence	L7, L40, L10

Table 5.2 *Synthesized comparison by strategy and granularity*

	Granularity	**Key Strategy**
L20	Passage	External knowledge retrieval
L21	Claim	External knowledge retrieval
L37	Passage	External knowledge retrieval
L41	Claims	External knowledge retrieval
L45	Sentence	External knowledge retrieval
L63	Claim	External knowledge retrieval
L12	Sentence	Self-consistency

L13	Sentence	Self-consistency
L26	Sentence	Self-consistency
L51	Sentence	Metamorphic-testing
L56	Passage	Metamorphic-testing
L36	Sentence	LLM-judge classifiers and ensembles
L44	Passage	LLM-judge classifiers and ensembles
L46	Sentence	LLM-judge classifiers and ensembles
L10	Sentence	LLM-judge classifiers and ensembles / LLM to train data
L7	Sentence	LLM to train data
L40	Sentence	LLM to train data

Table 5.3 *Comparison of LLM-as-a-judge hallucination detection methods*

5.1.6 Other techniques for knowledge-free black-box methods

Beyond LLM-as-a-judge and knowledge graphs, the literature presents several additional methodologies for hallucination detection that operate without access to model internals or external knowledge sources. These approaches, nine in total, can be categorized into four groups.

Reverse validation. This approach employs the principle of reverse validation, where the reliability of a generated answer is assessed by attempting to reconstruct the original query from the response. The reverse validation method operates on the assumption that if a language model generates a hallucination, it will struggle to consistently reconstruct the original query when the hallucinated answer is used as input. InterrogateLLM (L30), draws inspiration from human interrogation techniques where consistency in repeated questioning serves as an indicator of truthfulness. L2 extracts entities from the model's response and prompts the model to generate a question or a requirement based on that response. If the generated question matches the original user query, the response is deemed factual.

Uncertainty estimation. This approach detects hallucinations by estimating the uncertainty of the generated text. Given a query–answer pair, L5 paraphrases the query into multiple scenarios, compute a traditional uncertainty score for each paraphrase using uncertainty estimation method, and then apply factor analysis to disentangle a dominant common semantic factor from scenario-specific variation. L64 assumes that hallucinated tokens should receive systematically lower probability under a sufficiently strong proxy language model, and designs an uncertainty score by combining token-level negative log-likelihood and entropy, aggregated only over informative keywords such as entities and nouns.

Natural Language Inference (NLI). These techniques recast hallucination detection as a Natural Language Inference problem between the model's output and task-specific textual anchors. Using pretrained NLI models such as DeBERTa, RoBERTa, and BART, L34 performs zero-shot hallucination detection by checking entailment relations between source, target and hypothesis. L42 also relies on NLI, but focuses on internal consistency across multiple answers: it samples several responses for the same prompt, decomposes the top response into atomic claims, and for each claim uses an NLI model to measure how many alternative samples support, contradict, or remain neutral with respect to that claim.

Synthetic data. These approaches use synthetic data generation to train hallucination classifiers that can be applied at inference time. L17 introduces a perturbation-based pipeline in which a rewriting LLM takes system responses and produces paired faithful and hallucinated variants. These synthetic triples (context, faithful response, hallucinated response) are then used to fine-tune a detector. L35 generates large quantities of weakly supervised training data via LLM-based pseudo-labelling of unlabeled examples and rephrasings of labeled ones, then train an ensemble of DeBERTa-based classifiers. L43 trains a hallucination classifier by fine-tuning a RoBERTa model using Low-Rank Adaptation (LoRA).

5.2 Evaluation of hallucination detection methods

This section aims to answer the second research question in section 3.2: How have hallucination detection methods for LLM been evaluated? The study investigates benchmarks, metrics and experimental setups in the following subsections.

5.2.1 Evaluation benchmarks

Benchmarks are standardized datasets and structured tasks that enable to systematically assess the reliability, factuality, and faithfulness of model outputs. In the context of hallucination detection, benchmarks serve two primary purposes: quantify the tendency of LLMs to generate hallucinated content, and evaluate the effectiveness of detection methods in identifying such content.

Hallucination benchmarks are essential for establishing a unified framework to explore hallucinatory patterns in LLMs. These benchmarks are typically divided into hallucination evaluation benchmarks, which measure the prevalence of hallucinations in model outputs, and hallucination detection benchmarks, which assess the performance of detection algorithms across various domains and granularities, for example token, sentence or passage level [8].

In the researched studies, hallucination detection methods have been evaluated on a diverse group of datasets. Many benchmarks have been used to evaluate several different methods. The most used benchmarks are grouped in the Table 5.4.

Benchmark	**Size**	**Task**	**Metric**	**Used in**
HaluEval	5,000 / 30,000	Detection	Accuracy	L16, L17, L21, L29, L37, L45, L60, L63
SelfCheckGPT	1,908	Detection	AUROC	L5, L6, L16, L17, L32, L56, L60
SHROOM	3,000	Detection	Accuracy	L34, L35, L36, L40, L44
TruthfulQA	817	QA	LLM-judge, Human	L56, L57
FELM	3,948	Detection	Balanced Acc, F1	L21, L41
FreshQA	150	QA	Human	L41, L56
TriviaQA	95k	QA	Human	L28, L64
SQuAD	+100k	QA	Acc, P/R, F1	L28, L58
xSum	204,000	Summarization	ROUGE	L5, L19
SummEval	1,600	Summarization	ROUGE	L22, L23
PHD	100	Detection	P/R, F1	L2
SAC_3	250	Detection	AUROC	L12

Table 5.4 *Benchmarks used for evaluation in the studies.*

A major distinction could be found between general-purpose and task-specific benchmarks. General-purpose benchmarks evaluate hallucination detection across multiple tasks, domains, and model types: they are broad in scope and aim to provide a standardized evaluation framework for comparing different methods. Task-specific benchmarks evaluate hallucination detection in a particular task or domain, such as summarization or dialogue systems. They are often custom-built to answer the needs of a specific use case.

General-purpose benchmarks. The most used benchmark in this category is HaluEval, used for evaluation in L16, L17, L21, L29, L37, L45, L60 and L63. HaluEval is a large collection from year 2023 of 5,000 general user queries with ChatGPT responses and 30,000 task-specific examples from three tasks [11].

Task-specific benchmarks. Since this research excluded detection methods restricted to a specific domain or task, task-specific benchmarks were mostly not used in the studies. However, an exception can be considered SHROOM, a task-based hallucination detection dataset proposed as a task in 2024 as part of the 18th International Workshop on Semantic Evaluation [14].

Studies L34, L35, L36, L40 and L44 are all submissions to the SemEval-2024 task 6 and use SHROOM as a benchmark for the evaluation.

Baselines. In the context of hallucination detection, a baseline refers to a reference method or framework used for comparison when evaluating a new detection method. It serves as a benchmark, without being properly a dataset. The most used baseline is SelfCheckGPT (L13), used for evaluation in L5, L6, L16, L17, L32, L56 and L60. SelfCheckGPT is a zero-knowledge hallucination detection method, proposed in 2023. It works by asking the same model the same question multiple times and comparing the consistency of the responses. If the answers vary significantly, the original response is likely hallucinated [13]. The dataset associated with the method contains 1908 sentences from 298 articles generated by GPT-3 and labeled with one of the three veracity labels: accurate, minor inaccurate and major inaccurate. Fact-checking methods and tools are also used in several studies as baselines to evaluate the proposed methods. FacTool is used in studies L21, L41, L45 and L63 and FActScore in L18, L21 and L41.

5.2.2 Evaluation metrics

This section researches the most used evaluation metrics in the selected studies. Studies report different metrics, reflecting different needs (for example ranking or classification) or granularity (token, sentence, passage). The most frequent metric utilized in the studies are visualized in Figure 5.6.

Thresholded classification metrics. Under this category fall the metrics utilized to evaluate a binary decision (hallucination or not), after a threshold is chosen. Binary detectors that output a final hallucination binary label at claim, sentence, response, passage level, typically report these metrics.

Between these metrics, the most used in the studies are Accuracy, F1-score, Precision and Recall. Precision and Recall were always used together, as they describe two complementary aspects of classification performance, especially in tasks like hallucination detection, where the class imbalance can be great. The Balanced Accuracy metric, used in three studies, is also used to deal with imbalanced datasets.

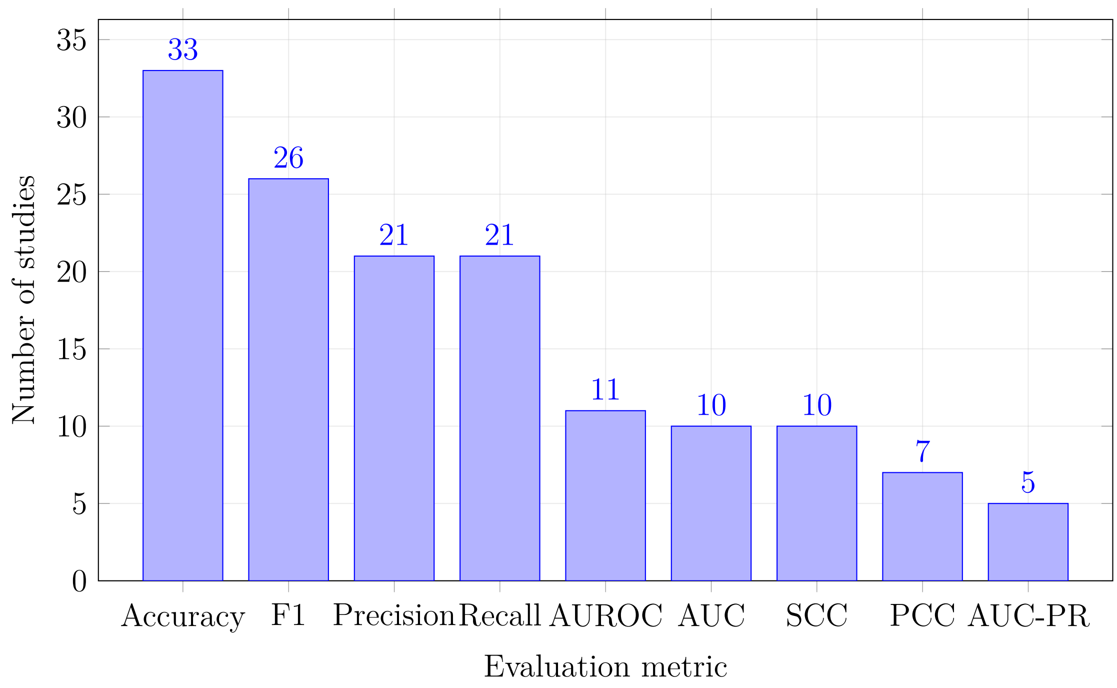

Figure 5.6 *Most used evaluation metrics in the selected studies.*

Ranking and score metrics. Under this category are grouped the evaluation metrics that output a continuous score, higher for hallucinations. These metrics returns a score and they don't need to be committed to a pre-decided threshold. The most frequent metrics in this category are Area Under Curve — Receiver Operating Characteristics (AUROC), Area Under Curve (AUC) and Area Under the Precision-Recall Curve(AUC-PR).

AUROC and AUC measure the model's ability to discriminate between hallucinated and non-hallucinated content across all possible classification thresholds. They represent the probability that a randomly chosen hallucination receives a higher score than a randomly chosen non-hallucination.

AUC-PR focuses on the trade-off between precision and recall, making it more informative when dealing with imbalanced datasets.

Correlation metrics. When a method outputs a continuous score, these metrics are used to measure how well these scores align with human-annotated scores. Pearson's correlation score (PCC) and Spearman's correlations score (SCC or ρ) fall under this category.

The Pearson correlation coefficient quantifies the strength and direction of a linear relationship between the model's predicted scores and the corresponding human ratings.

The Spearman correlation coefficient measures the monotonic relationship between two variables based on their rank order rather than their raw values.

These correlation metrics are relevant in sentence or passage level detection, where the goal is to rank outputs by their factuality or faithfulness rather than to make binary decisions.

6 Discussion

This systematic literature review examined hallucination detection methods for LLMs through the analysis of 50 studies published between 2023 and 2025. This chapter synthesizes the key results, interprets their significance, and discusses implications for both the research community and practitioners deploying LLMs in real-world applications.

Most of the studies were published in 2024. The 50 studies were firstly grouped by year: Figure 5.1 reveals that 2024 was the most prolific year. The literature review searched for studies from year 2018, although the first LLM, ChatGPT was released to the public on November 30, 2022. The results confirmed the fact that no studies were found previous to year 2022. The large amount of literature from year 2024 suggests that hallucination detection transitioned from a niche NLP concern to a critical reliability requirement following the widespread adoption of LLMs. The decline in 2025 papers within the selected scope, despite the technology's growth, can be explained with the fact that research focus is partially dedicated to closed-domain strategies (such as RAG) rather than pure detection in open-domain settings.

6.1 Hallucination detection methods

Majority of detection methods addressing factuality. Following the taxonomy of [8], the detection methods were classified into factuality or faithfulness hallucination detection strategies, as synthetized in section 2.3. Figure 5.2 shows a prominence of factuality hallucination detection (38 studies) over faithfulness detection (12 studies). This imbalance suggests that the primary concern in open-domain generation is the fabrication of world knowledge (entity errors, factual fabrications) rather than internal logical consistency. This focus is also intuitive given the open-domain nature of the selected studies: without grounding documents (as used in RAG, translation or summarization), the *truth* must be derived from the model's internal parametric knowledge or from external verification. However, the distinction is becoming blurred in knowledge graph -based approaches. These methods transform text into knowledge graphs or semantic graphs, treating factuality and faithfulness as the same mathematical problem: graph alignment or consistency.

Predominance of black-box methods. A significant finding of this review, visualized in Figure 5.3, is the majority of black-box methods (34 studies) over white-box methods (16 studies). This trend is probably a consequence of the commercial reality of the LLM landscape: as the most capable proprietary models (GPT, Claude, Gemini) are accessible primarily via API with no access to internal weights, logits, attention maps, gradients or hidden activations, researchers have been forced to innovate outside the model architecture. White-box methods generally offer finer granularity (token-level detection), but their detection is limited to open-source models (for example LLaMA and Mistral). By accessing the model's uncertainty directly, these methods also avoid the latency and computational expense of generating multiple external outputs. More capable proprietary models require also more computationally expensive (black-box) detection methods.

LLM-as-a-judge is the most used detection technique. Seventeen methods use LLM-as-a-judge approaches, where one language model evaluates the outputs of another. This evaluation strategy reflects both the capabilities and limitations of current AI systems. On one hand, LLMs possess the linguistic sophistication and reasoning ability to make nuanced judgments about factuality and consistency; on the other hand, LLM-as-a-judge methods inherit the very vulnerabilities they aim to detect. Evaluator models may themselves hallucinate, exhibit biases, or demonstrate inconsistent judgment. The literature attempts to mitigate this in different ways: grounding the judge in external evidence, querying multiple judges and using a majority vote, using self-consistency (ask the judge the same question under paraphrases or perturbations), combining LLM judge with other non LLM judgments or using LLM judges only to create supervision for separate detectors.

External knowledge detection in black-box methods. Among black-box methods, Figure 5.4 reveals 11 studies that rely on external references. External-knowledge methods align closely with the fact-checking paradigm described in section 2.3, typically combining fact extraction into atomic claims or triples and verification against web search, Wikipedia/Wikidata, or RDF knowledge graphs. These methods are particularly suited to domains where the relevant information is well covered by public knowledge bases. However, they are vulnerable to the same knowledge boundaries that affect the underlying LLMs: long-tail, up-to-date, or copyright-restricted knowledge remains difficult to verify. They also introduce engineering complexity and runtime cost due to retrieval and reasoning over external sources.

Reference-free detection in black-box methods. Zero-knowledge detection is attractive because it avoids dependencies on external infrastructure and can be applied in settings where retrieval is unavailable, unreliable, or undesirable. LLM-as-a-judge in this category is the most used technique (17 methods), along with knowledge graphs (7 methods). Moreover, other interesting minority approaches emerged: reverse validation, uncertainty estimation, Natural Language Inference and generation of synthetic data.

6.2 Hallucination detection methods evaluation

Broad diversity in evaluation benchmarks. As summarized in Table 5.4, studies rely on a variety of benchmarks, many of which differ substantially in task formulation, annotation protocol, size, and granularity. Some detectors are evaluated primarily on question answering datasets (TruthfulQA, FreshQA, TriviaQA and SQuAD), others on summarisation corpora (xSum, SummEval), and others on dedicated detection datasets like HaluEval, SelfCheckGPT or SHROOM. This diversity complicates systematic comparison across methods and reported performance is often tightly coupled to the characteristics of a particular benchmark. Further research could benefit from standardized benchmarks and evaluation protocols.

Evaluation metrics. As shown in Figure 5.6, most studies report standard classification metrics such as Accuracy, F1-score, Precision and Recall. Evaluation datasets are often imbalanced, explaining the use of Balanced Accuracy metric. Threshold-free ranking metrics such as AUROC, AUC and AUC-PR better describe the quality of the continuous scores produced by uncertainty-based and scoring-based methods. Finally, correlation metrics such as Pearson and Spearman are occasionally used to compare detector scores with human ratings.

6.3 Practical and theoretical implications

The 50 studies offer insights for researchers aiming to understand and standardize the field and for practitioners exploring the trade-offs of deploying LLM solutions in production.

Implications for practitioners without access to model internals. For practitioners relying on proprietary LLMs, the predominance of black-box methods validates the necessity of output-based detection strategies. These techniques often require multiple sampling iterations or external retrieval steps, introducing latency and increasing token costs, potentially making detection too cost-prohibitive for real-time consumer applications.

Feasibility for enterprise and safety-critical pipeline. Methods relying on external knowledge retrieval (fact-checking) offer high theoretical reliability for verifying facts but their reliability depends on the quality of the retrieval source and the long-tail knowledge problem. In contrast, zero-resource methods avoid external dependencies but often imply a high computational cost due to the need to generate multiple stochastic samples to measure inconsistency. The heavy reliance on LLM-as-a-judge poses a safety risk for critical pipelines. If the evaluator model shares the same biases or knowledge gaps as the generator, it may fail to detect subtle fabrications, creating a false sense of security. For safety-critical domains, human validation or hybrid approaches remain necessary.

Promising emerging approaches. Approaches that use large LLMs to generate synthetic training data to fine-tune smaller, specialized detection models (L7, L10, L40) offer an interesting path toward enterprise scalability. These methods effectively distill the reasoning capability of a large model into a cheaper, faster detector that can be deployed at inference time. Knowledge graph methods are significant because they treat factuality and faithfulness as a unified graph alignment divergence. By transforming text into graph structures, these methods treat hallucination detection as a graph alignment problem, potentially offering a unified mathematical approach for validating both real-world facts and internal consistency. Zero-resource methods that treat hallucination as model instability are promising because they operate without external resources. They are particularly valuable for environments where external retrieval is unreliable or unavailable.

Research gaps and methodological limitations. The usage of diverse, non-standardized benchmarks and inconsistent metrics makes it difficult to objectively compare methods with one another. Moreover, most methods output a binary classification (hallucinated or not hallucinated), missing a more nuanced classification.

6.4 Threats to Validity

This systematic literature review is subject to several threats to validity. The most important threat is the subjectivity associated with the single-reviewer process, potentially introducing subjective bias in study selection, quality assessment, and data extraction. Surely replication of inclusion and exclusion criteria and QA checklist by multiple reviewers would have enhanced the reliability of this study.

Search and selection bias. Several factors could have excluded important studies from the research. The exclusion of non-peer-reviewed literature (gray literature) ensures quality but may have excluded important developments, often published as preprints. Moreover, four otherwise relevant papers were unavailable via institutional access and could not be obtained from authors. An important threat is also the subjective bias associated with study selection. While many studies were clearly addressed at open-domain detection, the distinction was not always fully clear.

Data extraction and classification bias. Categorizing methods as factuality or faithfulness detection methods required judgment, especially for knowledge graph -based methods. These methods were classified as both factuality, faithfulness and both. Misclassification risk is mitigated by explicit taxonomies and a predefined extraction form, but residual subjectivity remains. Ambiguities also arise when methods support multiple granularities (token, claim, sentence, passage). In addition, some methods consisted of multiple main techniques, but for the purpose of categorization I assigned each to a single primary technique category.

7 Conclusion

This thesis aimed to systematically review hallucination detection methods for Large Language Models (LLMs) in open-domain settings, where no grounding documents are available. Hallucinations, outputs that are plausible yet incorrect or unfounded, pose significant risks to trust and usability. By synthesizing existing research, this work aimed to provide a structured overview of detection strategies, their evaluation practices, and emerging trends.

The review analyzed 50 peer-reviewed studies published between 2023 and 2025, revealing several key patterns. First, the research landscape is recent and rapidly evolving: no studies were found prior to 2022, and 2024 emerged as the most prolific year, reflecting the surge in interest following widespread LLM adoption. The decline in 2025 publications within the selected scope suggests a shift toward domain-specific approaches, such as Retrieval-Augmented Generation (RAG), rather than pure open-domain detection.

Second, the taxonomy proposed by Huang et al. (2025) [8] proved useful for classifying detection strategies into factuality and faithfulness hallucination detection. The majority of methods (38 out of 50) targeted factuality hallucinations (related to real world knowledge), while only 12 addressed faithfulness (adherence to user instructions and contextual consistency). This imbalance underscores the primary challenge in open-domain generation: verifying factual correctness without grounding documents.

A further distinction emerged between white-box and black-box approaches. Sixteen methods accessed model internals, enabling fine-grained uncertainty estimation and token-level detection. However, the dominance of black-box methods (34 studies) reflects a practical constraint: most used and powerful LLMs are proprietary and accessible only via API. Consequently, researchers have developed techniques that rely only on model outputs.

Among black-box strategies, two paradigms stood out. LLM-as-a-judge approaches, used in 17 studies, employ one model to evaluate another's outputs, sometimes combined with majority voting or external evidence. While promising, these methods inherit the vulnerabilities of LLMs themselves, including bias and inconsistency. Fact-checking methods represent another major category, typically decomposing outputs into atomic claims and verifying them against web sources, Wikipedia, or RDF knowledge graphs. These approaches offer strong performance when relevant knowledge is available but introduce complexity and remain sensitive to knowledge gaps.

Other noteworthy techniques include reverse validation, uncertainty estimation, Natural Language Inference (NLI)-based entailment, and synthetic data generation for training detectors. These methods highlight the creativity of the research community in addressing detection without relying on LLM internals or knowledge retrieval.

The second research question examined evaluation methodologies. The review found a lack of standardization: studies employed diverse benchmarks, metrics, and experimental setups. HaluEval was the most used general purpose benchmark, while SelfCheckGPT served both as a baseline method and dataset for detection. Metrics varied according to task formulation: classification-based methods reported Accuracy, F1, Precision, and Recall, while ranking-based approaches favored AUROC, AUC and AUC-PR. Correlation metrics such as Pearson and Spearman and PCC were used when aligning continuous scores with human judgments.

The findings reveal both progress and persistent challenges. Detection methods are diverse, in the goal to resolve the balance between scalability, accuracy, and independence from external resources. White-box approaches offer more precision but lack applicability to closed models. Black-box methods are broadly deployable but often computationally expensive or reliant on imperfect judges. Moreover, evaluation practices remain fragmented, limiting reproducibility and comparability.

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